
Haemophilia Intelligence Radar

Detects **what changed**, explains **why it matters**, names **who reviews it**, and proposes **what action is required**.

A news dashboard summarises documents.

This maintains state, and reports the transitions against it.

STATUS LEGEND

BUILT

PARTIAL

DESIGNED

NOT BUILT

— every capability in this deck carries one

If you read one slide, read this one

1,399

records under monitoring,
4 live public sources

134

forward-dated commitments
the Radar would be watching

3 of 15

golden questions the system
must *refuse* to answer

0

numbers in any answer that
came from a language model

- **We report change, not content.** Versioned state and a diff, so the output is `Phase 2 → Phase 3`, not "an article mentioned a trial".
- **We report absence.** A promised readout that never arrives is the signal. No document exists, so no summariser can find it.
- **We refuse.** A deterministic gate runs before generation; with thin evidence no model is called at all.

- **We get corrected.** Reviewer feedback is structured, not free text, so it can move scoring weights rather than just a record.
- **We state our limits.** Every capability on every slide carries a status chip, and what does not work is named on the slide where it would otherwise be claimed.

The hard part was never summarising.
It was noticing.

The cost is not missing information. It is late recognition.

Haemophilia intelligence arrives scattered across registries, regulators, journals, HTA bodies, congresses and investor pages — the same development, in different formats, on different days.

Consider one real shape of event:

WHAT ACTUALLY HAPPENS

A competitor's gene-therapy programme completes Phase 3 enrolment.

There is **no press release**. There is **no article**.

A single field in a public registry changes from ~~RECRUITING~~ to **ACTIVE_NOT_RECRUITING**.

There is nothing to summarise —
and that is exactly when it matters most.

WHY A SUMMARISER CANNOT HELP

- It answers *"what does this document say?"*
- The event is a **field change**, not a document
- The same item across three sources reads as three findings
- A promise that was never kept produces no document at all

WHAT IS ACTUALLY NEEDED

A system that already knows what it knew yesterday, and can therefore tell you what is different today — including the things that were supposed to be different and are not.

A news dashboard reports articles. The Radar reports state transitions.

WHAT A SUMMARISER PRODUCES — and what Citeline, Cortellis and Evaluate produce

"A recent article discusses ongoing developments in gene therapy for haemophilia B, noting that several competitor programmes are advancing through clinical stages."

ACTIONABLE?

No owner. No date. No claim you can check. Nothing a reviewer can argue with.

THE INCUMBENTS

Citeline, Cortellis and Evaluate are excellent *registries* — they will tell you a trial's status today. None of them tells you it **changed on Tuesday**, who owns that, or that a promised readout is now 41 days late.

WHAT THE RADAR PRODUCES

WHAT CHANGED

NCT0XXXXXXX ~~Phase 2~~ → **Phase 3**

overall_status · detected 12 Mar

WHY IT MATTERS

A competing asset advances in a reference geography, changing the near-term access picture for the same indication.

WHO REVIEWS

Clinical Development (primary) · Market Access (informed)

ACTION

LABEL_IMPACT_ASSESSMENT · SLA 7 d · NEW

EVIDENCE

registry record + exact source span, quoted inline

Monday, 09:10. A Market Access lead opens one queue — not a feed of articles, but **six cards that changed since Friday**, ranked. Three are already closed by a colleague. She reads the diff on the top one, disagrees with its urgency, clicks too early-stage, and reassigns it. That correction re-weights how the next comparable signal scores. Total time: four minutes.

One card per change, not one card per headline.

Every signal answers the same four questions, in the same order

HIGH

REGULATORY

HAEMOPHILIA B

SIG-0412

WHAT CHANGED

Regulatory status ~~FILED~~ → **APPROVED**

Geography EU · detected 12 Mar

WHY IT MATTERS

A competing asset gains market authorisation in a reference geography, changing the near-term access and positioning picture for the same indication.

WHO SHOULD REVIEW

Regulatory (primary) · Market Access (informed) — *reason shown, not implied*

ACTION

LABEL_IMPACT_ASSESSMENT

owner Regulatory · SLA 7 d · status NEW

EVIDENCE

EMA CHMP

COMPANY RELEASE

3 documents → 1 event

RENDERED, NOT WRITTEN

The **what changed** line is generated from the diff between two stored states. It is not free text from a model, so it cannot drift from what the source actually says.

PROVENANCE OR ABSTENTION

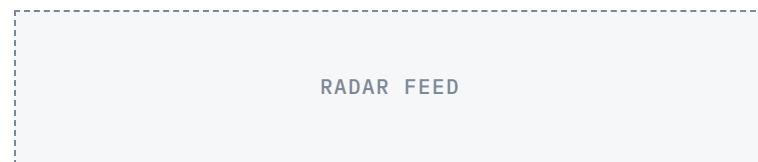
Every claim carries a source URL and an exact supporting quote. A claim that fails provenance validation is **withheld** — the card shows an explicit `insufficient evidence` state rather than a confident assertion.

AN ARTIFACT, NOT A NOTIFICATION

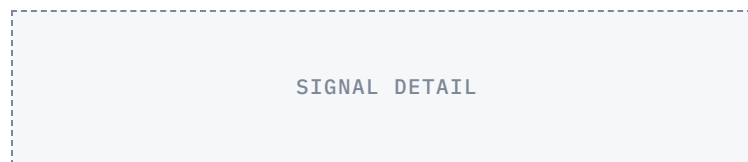
The card carries an owner, an SLA and a lifecycle (NEW → TRIAGED → ACTIONED → CLOSED). A reading task becomes a review-and-approve task, already assembled.

Three documents collapse into one event. Deduplication happens before a human ever sees it.

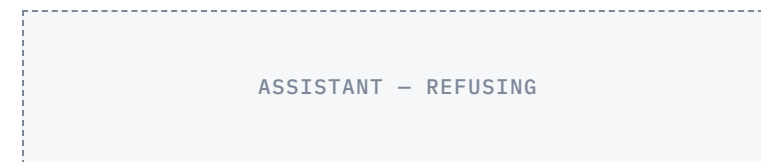
A working system against 1,399 real records — not a mock-up



Prioritised feed. Role-filtered queues, newest change first.



Diff panel and evidence drawer. Every claim expands to its source span.



The abstention path. The system declining a question it cannot support.

The demo includes a question the system refuses. Not an oversight — the capability we most want reviewed.

Three stages, two databases, one boundary that is enforced rather than agreed

Ingest	Dedup	Classify	Chunk + embed	Snapshot	Diff	Score + route	Signal + action
8 async connectors	idempotent upsert	rules, then LLM	FTS5 + ONNX	versioned state	old → new	urgency, owner	card, SLA



WHY A URI FLAG, NOT A CONVENTION

Phase 2 opens Phase 1's database as `mode=ro`. A stray write **raises** instead of silently corrupting.

In a system whose entire value is provenance, a bug in the reasoning layer must not be able to alter the evidence it reasons over. A team convention holds until someone is in a hurry at 2 a.m.

RETRIEVAL IS FOUR STAGES, NOT ONE

BM25 catches exact identifiers — `NCT06550882`, a sponsor's legal name — that embeddings blur. Vectors catch paraphrase BM25 misses. Fusion uses **ranks only**, so two incomparable scales need no tuning.

What is actually running — including what is not

SOURCE	RECORDS	STATUS
ClinicalTrials.gov v2	992	BUILT
PubMed / NCBI	299	BUILT
SEC EDGAR full-text	100	BUILT
arXiv	8	BUILT
OpenFDA	0	PARTIAL no matches
WHO ICTRP	0	PARTIAL blocks automation
medRxiv	0	PARTIAL no matches
USPTO	0	PARTIAL needs API key

All eight connectors are implemented and tested. Four returned nothing against live endpoints.

WE DO NOT RENDER EMPTY CATEGORIES

`signals.py` hard-codes the category map to the four sources that actually returned data, and the facets endpoint reports only categories genuinely present. Regulatory, HTA, Congress and Safety **have no source behind them in this build, so they are never emitted** — not shown as empty tabs.

CAPABILITY	STATUS
Versioned snapshots + change log	BUILT
Hybrid retrieval + evidence ranking	BUILT
Confidence gate / abstention	BUILT
Urgency scoring + role routing	BUILT
Feed / detail / assistant UI	BUILT
Classifier verdicts	PARTIAL rules live, LLM staged
Calibration re-weighting	PARTIAL captured, not wired
Expectation Ledger + silence	DESIGNED

Every one of the 1,399 records currently sits at "first seen". The diff engine is correct and has not yet had a second observation to diff against — a collection gap, not a code gap.

A promised readout is not news. It is a dated obligation.

Every system in this space treats "primary completion estimated October 2026" as text to be indexed.

It is not text. It is a **row in a ledger** — with a due date, an owner, and a quote that proves somebody said it on the record.

Then one of two things happens — and the second one is the whole point.

134 open commitments, from a field we already receive and currently throw away

LIVE COUNT • CLINICALTRIALS.GOV • 2026-09-07

Haemophilia studies matched	992
Carrying a primary completion date	946
Dated in the future	134
Typed ESTIMATED, not ACTUAL	134

Every single one is an **estimate** — a promise, not a fact. Among them: NCT07653139 (Novo Nordisk, 26 Oct), NCT06738485 (CSL Behring, 26 Oct), NCT06550882 (Takeda, 30 Sep).

THE SYSTEM ALREADY HALF-RECOGNISES THIS

```
# evidence.py:100 – trials held recency-neutral because
# ClinicalTrials.gov stores a date the study has not reached
if date_class not in ("publication", "filing"):
    return 1.00
```

That guard **is the detector**. It identifies exactly the records carrying a future date — and then discards the information. The Ledger is what that branch should feed.

SCHEMA

```
CREATE TABLE expectation (
  entity, asset, event_type,      -- who promised what
  window_start, window_end,      -- widened from `precision`
  precision,                      -- day|month|quarter|half|year
  quote_span NOT NULL,           -- no span, no row
  source_url NOT NULL,
  status,                          -- OPEN | WATCHING | MET
                                -- | MISSED | RETRACTED | SUPERSEDED
  next_check_at,                  -- drives the scheduler
  resolved_item_id, resolved_at
);
```

WATCH CADENCE – ONE ROW, FOUR PHASES

PHASE	WHEN	CADENCE
Dormant	< T-14d	weekly
Approach	T-14d → T	daily , targeted
Burst	T → T+grace	every 6 h
Resolve	> T+grace	silence signal

Reuses `scheduling_metadata.next_scheduled_run` — a column declared, documented, migrated, and **NULL for all 8 ingestors** because nothing ever wrote it.

The 134 registry commitments need no language model at all — date, sponsor, trial and event type are structured fields. Sprint one is field-mapping.

We do not claim nothing happened. We prove we looked.

SILENCEHIGHCLINICAL DEVELOPMENT

EXP-0088

WHAT CHANGED

Expected readout for NCT06550882 (Takeda) **did not arrive.**
promised window 2026-09-30 · grace 30 d · elapsed 41 d

WHY IT MATTERS

A slipped primary completion in a competing extended-half-life programme moves the comparative-data timeline in a reference market.

ACTION

HTA_WATCH_ITEM · owner Clinical Development · SLA 14 d

EVIDENCE

Original commitment: "Estimated primary completion date: 2026-09-30" (ClinicalTrials.gov, retrieved 2026-09-01)
63 probes across 4 sources between 2026-09-16 and 2026-11-10 — 0 matches. [probe log]

ABSENCE IS WHAT A MODEL WILL CONFABULATE

So the Ledger never asks one. Every scheduled probe is written to `watch_probe` whether it matches or not. **The log is the evidence.**

The claim is not *"we believe nothing happened"*. The claim is **"we ran 63 searches on these dates against these sources and found nothing"** — and a reviewer can open it.

WHY THIS IS UNUSUALLY DEFENSIBLE

- A reviewer can only disagree by **naming a source the probe did not cover** — which is a productive argument, and exactly what the calibration loop absorbs
- A completion date that slips twice is itself a signal. Supersession keeps the history rather than overwriting it

No article is ever written about a readout that did not happen.

The refusal is a feature, and it is enforced in code rather than asked for in a prompt

1 • THE GATE RUNS BEFORE GENERATION

With no evidence, evidence below `MIN_EVIDENCE`, or nothing but stubs, **no model is called at all**. A templated reply names what was missing.

A model asked to judge its own confidence is its own auditor, and it fails in one direction: it **under-refuses**, exactly when the corpus is thinnest.

supported	every claim traces to an excerpt or exact count
inferred	synthesised; no single excerpt says it
insufficient	the corpus cannot support an answer

2 • NUMBERS NEVER COME FROM THE MODEL

Top-k retrieval sees eight documents and **physically cannot** answer "how many trials are recruiting".

So the model **names a whitelisted query** from a fixed registry and supplies arguments. It never writes SQL and never sees a connection.

A generated query can be *plausible and wrong* — a missing `DISTINCT`, a silently dropped NULL — returning a confident number nobody can audit. In a regulated setting **a wrong number is worse than a refusal**.

3 • SELF-REPORT CAN ONLY LOWER

After generation, a `supported` claim with no citations — or citing records that were never retrieved — is **demoted**.

The asymmetry closes the last hole: a model may admit it is synthesising, and that is honoured. The reverse claim is not.

The `reason` string on every piece of evidence — "*matched on both keyword (#3) and meaning (#1, similarity 0.87)*" — is assembled **deterministically from the scoring fields**. No model writes it, so it cannot be fabricated.

Three of the fifteen golden questions must be refused. Abstention recall gates the suite's exit code — answering one of them fails the run even if every other answer is right.

A reviewer's correction changes the scoring, not just the record

BASELINE · BEFORE FEEDBACK

EVENT

Competitor gene-therapy programme completes Phase 3 enrolment (public registry update)

URGENCY

HIGH

ROUTE

Clinical Development

ACTION

Prepare competitive-response assessment · SLA 7 d

FEATURE WEIGHTS

enrolment milestone	0.40
readout proximity	0.10
HTA-relevant geography	0.15
indication overlap	0.35
sum	1.00

AFTER TWO REVIEWER CORRECTIONS

EVENT

Same event, re-scored

URGENCY

MEDIUM

ROUTE

Market Access & HEOR (primary) · Clinical Development (informed)

ACTION

HTA_WATCH_ITEM – expected readout registered as a silence watch · SLA 14 d

FEATURE WEIGHTS

enrolment milestone	0.15 ↓ .25
readout proximity	0.35 ↑ .25
HTA-relevant geography	0.25 ↑ .10
indication overlap	0.25 ↓ .10
sum	1.00

Reviewer: "Enrolment completion is already known and is not the decision point — the readout date is. The near-term implication for us is access, not trial design." **System:** the same four features re-weighted for this function — enrolment milestone -0.25, readout proximity +0.25, HTA geography +0.10, indication overlap -0.10. Nothing added or removed and **the weights still sum to 1.00**: a redistribution, not a new model.

Feedback is structured — already known · wrong indication · too early-stage · not our function — because free text cannot move a weight.

Five functions, five queues, five different definitions of urgent

FUNCTION	SEES	TYPICAL ACTION
Medical Affairs	evidence, congress, publications	EVIDENCE_REVIEW
Regulatory	filings, opinions, label changes	LABEL_IMPACT_ASSESSMENT
Market Access & HEOR	HTA, pricing, reimbursement	HTA_WATCH_ITEM
Clinical Development	trial design, endpoints, enrolment	TRIAL_DELTA_REVIEW
Commercial / Brand	competitor pipeline, corporate	POSITIONING_NOTE
Safety / PV	read-only visibility only	no case processing

Safety is deliberately **not** a routed queue. Anything touching adverse events is flagged for mandatory human review before it is shown as actionable, and the system never communicates externally.

URGENCY IS COMPOSED, NOT GUESSED	
FACTOR	WEIGHT
Trial changed status	×1.50
Tracked field moved	×1.25
Record first seen	×1.00
Completeness	0.60 – 1.00
Recency (publication/filing only)	0.90 – 1.10
Classifier verdict	1.00 – 1.10

THRESHOLDS CALIBRATED, NOT PICKED

With completeness capped at 1.00 and recency at 1.10, a never-changed record tops out at **1.07**. So `HIGH_THRESHOLD = 1.02` — any higher and the High band would be **unreachable until a second ingestion run**.

An empty tier is not a useful one. The same completeness and recency functions drive evidence ranking, so a record cannot be urgent here and unimportant there.

We measured it, including the questions it is supposed to fail

CATEGORY	Q	WHAT IT PROVES
factual_count	4	counts come from aggregates, not generation
semantic_search	2	paraphrase retrieval works without keywords
source_specific	1	source filters are honoured
insufficient_evidence	3	the system refuses rather than guesses
cross_document	2	synthesis is labelled <code>inferred</code> , not <code>supported</code>
citation_correctness	1	every citation resolves to a retrieved record
item_scope	2	item chat cannot reach outside its record

ASSERTION VOCABULARY

`expect_route` · `expect_numbers` · `expect_contains`
`expect_absent` · `expect_confidence` · `must_retrieve_any_of`
`assert_citations_valid` · `scope`

Assertions are declarative in `eval/golden.yaml`, so adding a regression case is a YAML entry, not code.

Expected values are corpus-specific constants. Re-ingesting breaks the suite loudly rather than letting it silently pass — which is the point.

RESULTS TABLE

per-category pass rate + abstention recall
from `scripts/run_eval.py`

ABSTENTION RECALL GATES THE EXIT CODE

It is reported **separately** from accuracy, and a recall below 1.0 fails the run. A system that answers a question it cannot support has failed, regardless of how many other answers were right.

WHAT WE DELIBERATELY DO NOT CLAIM TO HAVE MEASURED

- Extraction precision/recall, dedup cluster purity, diff correctness — the single-backfill corpus produces no diffs and no duplicate clusters to score
- Precision@10 before vs after calibration — feedback is captured, the re-weighting loop is not yet wired

What earlier recognition is worth, with the arithmetic shown

THE DECISIONS THAT CHANGE

- **An HTA rejection in a reference market**, recognised ten days earlier, gives Market Access a response window *before* follow-on national assessments anchor to it
- **A competitor readout that slips** is a planning input available nowhere else — today nobody notices it at all, because no document is published when a date is missed
- **A label change caught at filing** rather than at approval moves the comparative-evidence conversation upstream of the launch plan

WHY WE ARE NOT QUOTING A CURRENCY FIGURE

Converting these into money needs internal cost and portfolio-value inputs we do not have and would have to invent. We would rather show a model you can argue with than a number you cannot check.

TIME SAVED – THE WEAKER HALF OF THE CASE

ASSUMPTION	VALUE
Competitor programmes tracked	40
Functions consuming intelligence	5
Manual scanning, per function	6 h/wk
Residual review of assembled cards	8 h/wk
Lag today → with monitoring	11 d → 1 d
Scanning load today 5 × 6	30 h/wk
Time redirected	22 h/wk ≈ 0.55 FTE
Recognition lag reduced	~10 days

Sensitivity — half the assumed load: 0.27 FTE. Double: 1.1 FTE. **The lag figure is insensitive to all of them**, which is why it, not the FTE count, is the number we stand behind.

Assumptions are ours, not Novo Nordisk's. Argue with the inputs — the arithmetic is on the slide.

Cheap to run, public by construction, and the next sprint is already specified

OPERATING PROFILE	
Embeddings	local, £0
Model weights on disk	90 MB
Corpus	4.4 MB
GPU required	none
Data sources	public only
Internal data required	none
Inference is the only recurring cost, and only for summaries and chat. Retrieval, scoring, routing, diffing and the entire Expectation Ledger run with no model calls at all .	

DEPLOYMENT POSTURE	
• SQLite ships in the repo — a reviewer clones and runs; no server to provision • Schema is portable; Postgres + pgvector is the documented migration when corpus size demands it (ADR 001) • The model is a config value behind one interface — swapping providers is a re-baseline of the eval suite, not a rewrite • No login, no telemetry, no third-party analytics	

NEXT 90 DAYS	
Sprint 1	Expectation Ledger — registry path only. Keep <code>primaryCompletionDateStruct</code> , migration <code>003</code> , date triggers, probe log. No LLM needed.
Sprint 2	Re-run ingestion on schedule → first real diffs; re-baseline the eval constants
Sprint 3	Wire the calibration loop; measure precision@10 before vs after
Sprint 4	Enrichment for <code>retracted</code> / <code>citation_count</code> ; extend the Ledger to prose sources

The differentiating capability is the cheapest thing left to build. Sprint 1 is field-mapping and a scheduler trigger — roughly 330 lines, and no model call.

What we need from you — and what we are not claiming

WHAT WE WOULD NEED FROM NOVO NORDISK

- One or two stakeholders per function for **two short calibration sessions** — the feedback loop is only demonstrable against real reviewer judgement
- Confirmation of **which functions are routing targets**, and whether Safety/PV is a routed queue or read-only visibility
- **Urgency definitions and SLA expectations per function**, so scoring is anchored to real practice rather than assumed
- **Priority geographies** for HTA and regulatory weighting
- Sources to **add to, or exclude from**, the allowlist
- Guidance on phrasing competitive impact, given the prototype references Novo Nordisk assets only from public information

THE THREE THINGS WE ARE NOT CLAIMING

- The **Expectation Ledger has never run**. The 134 is a live query against ClinicalTrials.gov, not our output
- No **real diff** has occurred yet — the corpus is a single backfill
- The **calibration improvement is unmeasured**; feedback is captured, the re-weighting loop is not wired

The full list is in `LIMITATIONS.md` and the risk register in the appendix. Nothing in this deck is claimed without a status chip.

WHERE WE WOULD START

Sprint one is the Expectation Ledger on the registry path: keep one field the API already returns, add one migration, swap an interval trigger for a date trigger. **~330 lines, no language model, and the 134 rows are already there.**

A system that tells you what it does not know is the only kind you can act on.

Appendix

Source allowlist and counts · schemas · endpoint inventory · scoring weights ·
decision records · risk register.

Source allowlist, access terms, and what each returned

SOURCE	ENDPOINT	RECORDS	AUTH	TERMS
ClinicalTrials.gov v2	/api/v2/studies	992	none	US Gov, public domain
PubMed / NCBI	esearch + efetch	299	none	NCBI usage policy
SEC EDGAR	efts.sec.gov/LATEST	100	none	Fair access; identifying UA sent
arXiv	export.arxiv.org/api	8	none	arXiv API terms, metadata only
OpenFDA	api.fda.gov/drug/*	0	none	US Gov, public domain
WHO ICTRP	trialsearch.who.int	0	none	Blocks automated access
medRxiv	api.biorxiv.org	0	none	Metadata only
USPTO	developer.uspto.gov	0	API key	Key not held

TRIAL SET • TOP SPONSORS	
Novo Nordisk A/S	105
Bayer	58
Baxalta / Shire	54
Pfizer	43
CSL Behring	29

TRIAL SET • STATUS DISTRIBUTION			
COMPLETED	588	UNKNOWN	124
RECRUITING	90	TERMINATED	71
ACTIVE_NOT_REC.	44	NOT_YET_REC.	38
Only RECRUITING and TERMINATED map to deterministic verdicts today — 161 of 1,399.			

Schemas and API surface

PHASE 1 • HAEMOPHILIA_DATA.DB

raw_items	id • source_name • source_id source_url • title • raw_content date_published • date_ingested metadata (JSON) • is_active
ingestor_logs	run history, per connector
scheduling_metadata	cursor • last_run next_scheduled_run ← never written
classifications	raw_item_id • category • justification method • status • reason • classified_at

PHASE 2 • PHASE2.DB (SIDECAR, ALL WRITES)

item • chunk • chunk_fts • embedding
item_snapshot • item_change • item_review
chat_thread • chat_message
landscape • summary_cache

API • 18 ENDPOINTS

Health	/api/health
Feed	/api/feed • /api/facets /api/landscape • /api/landscape/refresh
Items	/api/items/{id} /api/items/{id}/summary /api/items/{id}/similar
Signals	/api/signals • /api/signals/{id} /api/signals/{id}/review
Chat	/api/chat/item/{id} /api/chat/global • /api/threads/{id}
Papers	/api/papers • /api/papers/{id} /api/categories/{cat}
Classification	/api/classifications/recheck

Both chat endpoints stream over SSE with a **withholding buffer**, so a reply that fails post-generation validation is never partially shown and then retracted.

Decision records — and what each one cost us

#	DECISION	REJECTED	COST ACCEPTED
001	SQLite + FTS5	PostgreSQL + pgvector	No concurrent writers; vector search is a full scan
002	fastembed ONNX	sentence-transformers	Smaller model retrieves worse; 90 MB instead of 2.5 GB
003	Phase 1 frozen, <code>mode=ro</code>	One shared schema	Derived state duplicated in a sidecar
004	Model names a whitelisted query	Text-to-SQL	Only anticipated aggregates are answerable
005	One tool round-trip, never two	Open agent loop	Multi-hop questions cannot be answered
006	Confidence gate before generation	Prompt-based "say I don't know"	Refuses some questions a human could answer
007	Reciprocal Rank Fusion	Weighted score fusion	Score magnitude discarded
008	Trials held recency-neutral	Uniform recency	Genuinely recent trials get no boost

Every record names the alternative that was rejected and what it cost. A decision with no stated cost was not a decision.

Risk register

RISK	IMPACT	MITIGATION IN THIS BUILD	RESIDUAL
Source changes API or blocks automation	High	Isolated per-connector; failures logged, run continues	Realised — WHO ICTRP
Rate limiting under production volume	High	Backoff, per-source intervals, <code>max_instances=1</code>	Untested above backfill volume
Model fabricates a claim or number	Critical	Aggregates never generated (ADR 004); provenance-or-withhold; gate before generation	Reasoning prose between cited facts
Silence signal asserts a false absence	High	Every probe logged; card cites the log, not a belief	A gap in the allowlist looks like silence
Corpus staleness	High	Scheduler built, cursors resumable	Realised — single backfill
Verdicts rest on uncollected fields	Medium	Verdict weight capped 1.00–1.10 so it cannot dominate	Realised — one category unreachable
Sensitive signal actioned without review	Critical	Mandatory human review; no external communication	Policy and UI, not a hard control
Eval constants break on re-ingestion	Low	Suite fails loudly rather than passing silently	Re-baseline required each run